

ASSIGNMENT 1

Course	Full Stack Web Development
Course Code	23CM4121
Assignment No.	1
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Section	B

1. TITLE

Implement Inheritance Types using JavaScript

2. AIM

To implement and demonstrate different types of inheritance in JavaScript using ES6 classes, class extension, and mixins.

3. OBJECTIVES

1. To understand inheritance in JavaScript.
2. To implement Single Inheritance.
3. To implement Multilevel Inheritance.
4. To implement Hierarchical Inheritance.
5. To demonstrate Multiple Inheritance using Mixins.

4. THEORY

Inheritance is an object-oriented programming concept in which a class can acquire properties and methods from another class. In JavaScript, inheritance is commonly implemented using the **extends** keyword and **super()** constructor. JavaScript does not support multiple inheritance of classes directly, so multiple behaviors can be combined using mixins.

5. ALGORITHM / PROCEDURE

1. Create a JavaScript file.
2. Define a base class and a derived class for Single Inheritance.
3. Create a chain of classes for Multilevel Inheritance.
4. Create multiple child classes from one parent class for Hierarchical Inheritance.
5. Create mixin objects and combine their methods with a class prototype for Multiple Inheritance.
6. Create objects and call the inherited and own methods.
7. Save and execute the file.
8. Verify the output.

6. PROGRAM

```
Welcome x JS Inheritance.js
C:\Users\DELL\Desktop> JS Inheritance.js > ...
1 //Single Inheritance
2 console.log("Single Inheritance");
3 class Animal {
4     constructor(name){
5         this.name=name;
6     }
7     eat(){
8         console.log(`${this.name} is eating.`);
9     }
10 }
11 class Dog extends Animal{
12     bark(){
13         console.log(`${this.name} is barking.`)
14     }
15 }
16 const myDog = new Dog("Buddy");
17 myDog.eat();
18 myDog.bark();
19
20 //Multilevel Inheritance
21 console.log("Multilevel Inheritance");
22 class Vehicle {
23     move(){
24         console.log("Moving forward");
25     }
26 }
27 class Bike extends Vehicle{
28     wheels(){
29         console.log("Bike has 2 Wheels");
30     }
31 }
32 class Car extends Bike{
33     windows(){
34         console.log("Car has 4 Windows");
35     }
36 }
37 const v = new Car();
38 v.move();
39 v.wheels();
40 v.windows();
41
42 //Hierarchial Inheritance
43 console.log("Hierarchial Inheritance");
44 class User{
45     constructor(username){
46         this.username = username;
47     }
48     login(){
49         console.log(`${this.username} logged in.`);
50     }
51 }
52 class Admin extends User{
53     deleteUser(){
54         console.log(`${this.username} deleted by admin`);
55     }
56 }
57 class Customer extends User{
58     checkout(){
59         console.log(`${this.username} checked out successfully.`);
60     }
61 }
62 const a = new Admin("Anvitha");
63 const c = new Customer("Sam");
64 a.login();
65 c.login();
66 a.deleteUser();
67 c.checkout();
68
```

```
Welcome X JS Inheritance.js
C: > Users > DELL > Desktop > JS Inheritance.js > ...
68
69 //Multiple Inheritance
70 console.log("Multiple Inheritance");
71 //Using Mixins
72 const hall = {
73   sofa(){
74     console.log("Sofa is placed in Hall.");
75   },
76   tv(){
77     console.log("Tv is placed in Hall.")
78   }
79 };
80 const bedRoom = {
81   bed(){
82     console.log("Bed is in bedroom.");
83   },
84   closet(){
85     console.log("Closet is in Bedroom.");
86   }
87 };
88 class Building{
89   constructor(houseNo){
90     this.houseNo = houseNo;
91   }
92   house(){
93     console.log(`Building House No: ${this.houseNo}`);
94   }
95 }
96 class Area extends Building{
97   constructor(houseNo,area){
98     super(houseNo);
99     this.area = area;
100   }
101 }
102 Object.assign(Area.prototype,hall,bedRoom);
103 const name = new Area("403","Kommadi");
104 name.house();
105 name.sofa();
106 name.bed();
```

7. CODE EXPLANATION

Inheritance Type	Implementation
Single Inheritance	Dog extends Animal. Dog inherits eat() and adds bark().
Multilevel Inheritance	Car extends Bike, and Bike extends Vehicle.
Hierarchical Inheritance	Admin and Customer both extend User.
Multiple Inheritance	Mixins are combined with Area using Object.assign().

8.OUTPUT

The following output is obtained after executing the program:

```
PS C:\Users\DELL\Desktop\Assignment> cd ..
PS C:\Users\DELL\Desktop> node Assignment.js
Single Inheritance
Buddy is eating.
Buddy is barking.
Multilevel Inheritance
Moving forward
Bike has 2 Wheels
Car has 4 Windows
Hierarchial Inheritance
Anvitha logged in.
Sam logged in.
Anvitha deleted by admin
Sam checked out successfully.
Multiple Inheritance
Building House No: 403
Sofa is placed in Hall.
Bed is in bedroom.
PS C:\Users\DELL\Desktop> 
```

8. RESULT

Thus, the different types of inheritance—Single, Multilevel, Hierarchical, and Multiple Inheritance using Mixins—were successfully implemented using JavaScript.